

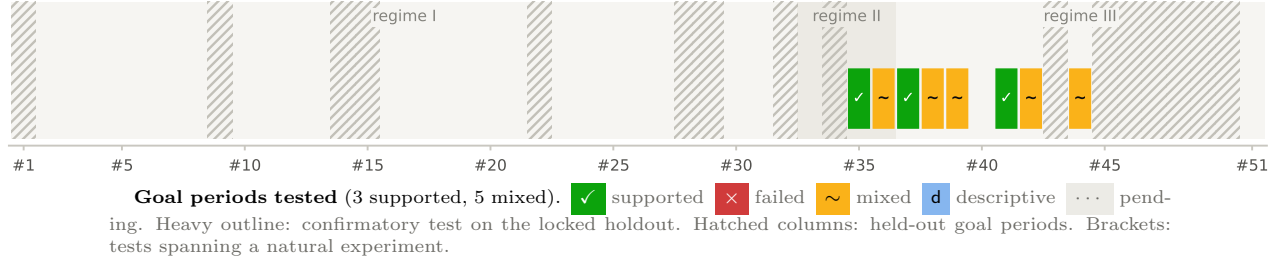
H100: #best/#rest: explicit vs spontaneous symmetry breaking

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 45% Faithfulness 2/5 Usefulness 2/5

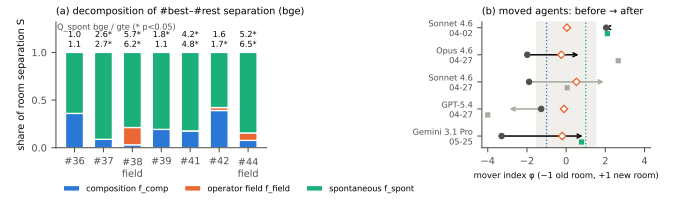
Bottom line. The #best/#rest content split is mostly endogenous and regenerates at each goal; the operator's room instruction triggers a larger split but not along its own text, and composition explains little.

level: hypothesis · pipeline checks 100% · updated 2026-10-04



WHAT HAPPENED

HH66 says the #best/#rest content difference has two parts: a response to the operator's room field (explicit symmetry breaking) and a spontaneous divergence that stays with the room when members swap. H100 decomposes the room separation in 8 non-holdout two-room periods (#35–#44) into composition, field and spontaneous shares. It also follows 5 operator moves. **The difference is endogenous.** After the members' own constants and the measured field directions are removed, the rooms still differ beyond random groupings in 5/7 regime-III periods (f_{spont} 0.58–0.91). Composition is small. The kickoff-text direction carries a chance share, although the two fielded weeks separate most. The divergence does not survive the next goal. One mover adopts its new room within a day, and one keeps its old position all week. The holdout (NE19) is not run.



Observables. (a) Shares of the room separation per regime-III period (bge). Numbers give Q_{spont} for bge and gte (* $p < 0.05$). Composition (blue) and the operator-field direction (orange) are small; the endogenous part (green) dominates. (b) Mover index before → after each operator move. Grey arrows: the post-move native contrast is not significant. Diamonds: composition prediction. Squares: gte.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

For an operator, rooms with identical instructions still diverge (Q_{spont} 1.8–4.2 in 3/5 such weeks). Different instructions about double the separation (Q 5–7), but the content does not line up with the instruction text, so a text-similarity monitor of “is each room on its brief” will miss it. Room identity does not carry over a goal change: regroup freely, nothing persists. A moved agent may align with its new room on day 1 or keep its old project for a week. Check its work, not its room. This is a modest design fact. It says where room differences come from, not how to steer them.

PHYSICAL MODEL

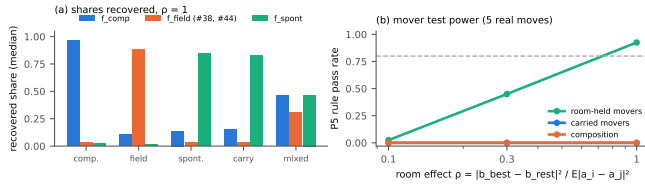
Model 11 (soft-spin $O(n)$, $n = 32$ whitened dimensions) with the room as a Potts domain index (model 10). The spin $\mathbf{x}_{i,d}$ is agent i 's style-residualized content on day d . $\mathbf{g}_d$ is the day field (goal, schedule), $\mathbf{a}_i$ the agent constant, $\mathbf{h}_r$ the operator room field (room kickoff, operator messages) with susceptibility χ , and $\mathbf{m}_r$ the spontaneous room magnetization. S is the split-half room separation and Q its excess over random partitions. The mover index φ is +1 for a native of the new room.

$$\begin{aligned}\mathbf{x}_{i,d} &= \mathbf{g}_d + \mathbf{a}_i + \chi \mathbf{h}_{r(i,d)} + \mathbf{m}_{r(i,d)} + \boldsymbol{\epsilon}_{i,d} \\ S &= \Delta^{(1)} \cdot \Delta^{(2)}, \quad Q = S / \langle S \rangle_{\text{relabel}} \\ f_{\text{comp}} &= \Delta_{\hat{a}} \cdot \Delta / S, \quad f_{\text{field}} = \sum_k (\Delta^{(1)} \cdot \mathbf{u}_k) (\Delta^{(2)} \cdot \mathbf{u}_k) / S \\ \varphi_k &= [\text{sim}(k, \text{new}) - \text{sim}(k, \text{old})] / C_{\text{native}}\end{aligned}$$

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RESULTS VS PREDICTIONS

Prediction (locked)	Observed	Verdict
P0 synthetic: size, shares, mover power	Q size 0.04; shares separate mostly worlds; power 0.93 at $\rho = 1$, 0.45 at 0.3	
P1 $Q > 1$ in $\geq 6/8$	bge 5/8, gte 6/8	mixed
P2 $f_{\text{comp}} \leq 0.3$	0.03–0.18 where rooms separate; supported constants invariant (r 0.49 vs 0.14)	
P3 field share ≥ 0.15 in #38, #44	0.18 (null 0.16), 0.08 (null 0.03); failed gte 0.00, 0.04	
P4 spontaneous, identical kickoffs	$Q_{\text{spont}} p < 0.05$ in 3/5; #41 4.23 supported (p 0.001)	
P5 movers adopt new room	3/5 bge, 2/5 gte; GPT-5.4 carries mixed ($\varphi -3.0$)	
P6 no remanence	$R(\#39, \#41) z -1.6$; all pairs supported $ z < 2$	
P7 swap-carry null	$R_{\text{room}} -0.23, R_{\text{agent}} -0.21$ (n.s.) supported	
Post hoc: fielded weeks largest	Q 6.85, 5.09 vs median 1.68 (rank lead p 0.048)	



Second figure. Synthetic validation on the real agent-day skeleton (40 worlds per setting). (a) Recovered shares at $\rho = 1$ in the composition, field, spontaneous, carried-mover and mixed worlds. (b) Pass rate of the mover rule against room-effect size; carried and composition worlds never pass.

GOAL PERIODS IN DETAIL

G35 ✓ supported
 G36 ~ mixed
 G37 ✓ supported
 G38 ~ mixed
 G39 ~ mixed
 G41 ✓ supported
 G42 ~ mixed
 G44 ~ mixed

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writeup/paper.tex`).

CAVEATS

Five movers, all at goal starts; at 04-27 the native contrast is marginal (p 0.04–0.10) and the mover test has power 0.45 at $\rho = 0.3$. The field is measured as the whitened kickoff-text direction, so a failed P3 does not say the instruction had no effect. The endogenous share includes room drives the members create, such as shared repos and the #35 forks; coupling through reading is not separated (convergence impostor open). Periods are 3–17 days. The ranking of fielded weeks and the per-mover narratives are post hoc.

ROUND 2

Confirm on NE19 (#45). Measure the field as DQ4 work per room instead of kickoff text. Test movers' drift toward read vs not-yet-read new-room messages at matched age.

- **Essence.** whether a room is an order parameter of its own (a domain that holds content independent of its members and its instructions), and what an operator's room assignment does to content.
- **H100-R1** Confirm on NE19 (confirm.py C1) and the held-out two-room weeks.
- **H100-R2** Read-out mechanism for movers.
- **H100-R3** The field as work, not text.
- **H100-R4** GPT-5.4's carry and Sonnet 4.6's pre-alignment.
- **H100-R5** Coordination with H81–H83 (composition, remanence, enculturation): the same agent constants and the same "no remanence" reading at village level.

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